

Research Grounding and Build Notes

Your requested convergence is technically actionable when treated as a **search-and-evaluation system**, not a single all-purpose prompt.

Key evidence

- Gu et al., *Generating Creative Ideas via LLMs for Scientific Research* (2025) implement a two-phase pattern: knowledge preparation, then combinatorial generation. Their retrieval represents innovations at multiple abstraction levels and their process decomposes components before integrating them. The paper reports stronger alignment with later research developments than its direct-generation baseline.
- The same paper uses Boden's distinction: combinatorial creativity recombines familiar ideas; exploratory creativity searches a conceptual space; transformational creativity changes its rules. The MVP begins at combinatorial creativity and leaves clear hooks for the latter two.
- Zhu, *The Internet of Agentic AI* (2026) frames networked agents as systems of planning, tool use, communication, coordination, trust, and governance. Its useful concept here is **controlled emergence**: allow novel collective behavior while bounding it with protocol, resource, identity, and governance layers.

Practical decisions

1. **Start with a narrow arena.** Pick one mission class—e.g., climate adaptation projects, developer tools, community health operations, or education—not “all ideas.” A sharp evaluation function is an accelerator, not a limitation.
2. **Store mechanisms and constraints.** Plain documents and embeddings are insufficient for creative transfer. Capture what a concept does, when it fails, and at least four abstraction levels.
3. **Separate four passes.** Retrieve → decompose → bridge → criticize. Combining them in one prompt encourages fluent but untested analogy.
4. **Make disagreement an artifact.** The critic must name what would falsify the bridge and which stakeholder might be harmed or excluded.
5. **Close the loop cheaply.** Prototype with a simulation, code test, paper prototype, or small user study. Save metrics and decisions back into memory.
6. **Use evolutionary selection only after measurement exists.** Keep a population of candidates, mutate/recombine high-scoring ones, but base survival on observed outcomes rather than model self-grading alone.

30-day plan

- Days 1–3: Choose mission, user/stakeholder, constraints, and one measurable outcome. Add 30 curated atoms across 3 domains.
- Days 4–7: Run 20 challenges through the included engine. Human-rate bridge quality, feasibility, and usefulness; revise the atom format.
- Week 2: Add an LLM adapter that produces structured JSON under the same card schema. Retain deterministic retrieval and the critic.
- Week 3: Add source-backed web/library ingestion plus a small graph database. Introduce provenance and contradiction records.
- Week 4: Build a protected prototype runner and a human review UI. Trial one reversible candidate with real users or a simulator.

Sources

- Gu, T. et al. (2025). “Generating Creative Ideas via LLMs for Scientific Research.” arXiv:2412.14141. <https://arxiv.org/abs/2412.14141>
- Zhu, Q. (2026). “The Internet of Agentic AI: Communication, Coordination, and Collective Intelligence at Scale.” arXiv:2606.12835. <https://arxiv.org/abs/2606.12835>
- Boden, M. A. (2004). *The Creative Mind: Myths and Mechanisms*, 2nd ed. Routledge.